A Data-Driven Assessment of Transfer Efficiency, Care Backlogs, and Placement Outcomes

1. Executive Summary

The Unaccompanied Alien Children (UAC) Program operates as a multi-stage care and reunification pipeline involving apprehension and CBP custody, transfer to HHS care, medical screening, sheltering, case management, and eventual discharge to a vetted sponsor. The effectiveness of this system depends not only on the capacity available at each stage but also on how efficiently children move through the pipeline and reach successful placement outcomes.

This project, **Care Transition Efficiency & Placement Outcome Analytics**, analyzes operational reporting data to evaluate transition efficiency, discharge performance, backlog pressure, temporal patterns, and outcome stability.

The analysis models the process as:

CBP Custody → HHS Care → Sponsor Placement

Three primary transition efficiency metrics are evaluated:

- **Transfer Efficiency Ratio** = Transfers ÷ CBP Custody
- **Discharge Effectiveness** = Discharges ÷ HHS Care
- **Pipeline Throughput Rate** = Total Exits ÷ Total Entries

Additional analysis examines the relationship between inflows and successful exits to identify potential accumulation pressure, sustained imbalance periods, and prolonged low-discharge periods.

Temporal analysis evaluates weekday versus weekend patterns, month-over-month placement trends, and periods of potential stagnation. Outcome stability is assessed using discharge variability, rolling performance measures, and detection of unusually large declines in discharge activity.

The project also develops an interactive **Streamlit dashboard** containing four core modules:

1. Care Pipeline Flow Visualization
2. Transfer & Discharge Efficiency Panels
3. Bottleneck Detection Charts
4. Outcome Trend Analysis

The resulting analytical framework shifts the focus from simply monitoring the number of children present in care toward evaluating **process efficiency, transition performance, bottlenecks, and placement outcomes**

2. Introduction

2.1 Background

The Unaccompanied Alien Children (UAC) Program is a multi-stage care and reunification system. Children initially enter the process through apprehension and placement in CBP custody. They are subsequently transferred to the care of the Department of Health and Human Services (HHS), where they receive services including medical screening, sheltering, and case management. The process ultimately aims to discharge children to a vetted sponsor.

The operational structure can therefore be represented as a pipeline:

Apprehension

↓

CBP Custody

↓

Transfer to HHS

↓

HHS Care

↓

Case Management

↓

Sponsor Placement

The project focuses particularly on the transition stages represented in the available reporting data:

CBP Custody

↓

HHS Care

↓

Sponsor Placement

This pipeline perspective is important because system performance cannot be evaluated solely by examining the number of children present in custody or care. A system may have substantial capacity while still experiencing delays if children are not moving efficiently between stages.

3. Problem Statement

The project addresses the need to evaluate how efficiently children move through the UAC care pipeline and whether the system is maintaining a sustainable balance between incoming cases and successful exits.

The primary analytical questions are:

1. **How efficiently are children transferred from CBP custody to HHS care?**
2. **Are HHS discharges keeping pace with incoming transfers?**
3. **When does potential backlog accumulation occur?**
4. **Where and when do bottlenecks emerge?**
5. **Are placement outcomes stable over time?**
6. **Are there periods of unusually low discharge activity?**

The project guide specifically frames these questions around transition efficiency, discharge and sponsor placement outcomes, and identification of delays and bottlenecks.

4. Objectives

4.1 Primary Objectives

The primary objectives of the project are:

Objective 1 — Measure CBP → HHS Transition Efficiency

Evaluate the relationship between children transferred from CBP custody and the number of children in CBP custody.

Objective 2 — Evaluate Discharge and Sponsor Placement Outcomes

Analyze the volume and effectiveness of children discharged from HHS care.

Objective 3 — Identify Delays and Bottlenecks

Compare inflows and successful exits to identify periods of imbalance and potential accumulation pressure.

4.2 Secondary Objectives

The project also aims to:

- Support faster reunification and sponsor placement.
- Identify potential opportunities for improving case-management workflows.
- Identify periods of operational instability.
- Provide analytical evidence that can support process-level improvements.
- Develop an interactive dashboard for monitoring transition and placement performance.

5. Dataset Description

The analysis uses reporting data containing information about children entering, moving through, and exiting the UAC care pipeline.

The principal variables are:

Variable	Description	Analytical Role
Date	Reporting date	Temporal analysis
Children apprehended and placed in CBP custody*	Daily intake volume	Pipeline entry

Variable	Description	Analytical Role
Children in CBP custody	Children currently in CBP custody	CBP stock
Children transferred out of CBP custody	Children transferred into HHS care	Pipeline flow
Children in HHS Care	Children currently in HHS care	HHS stock
Children discharged from HHS Care	Children discharged to sponsors	Pipeline exit / placement outcome

6. Analytical Framework

The project models the UAC system as a **flow pipeline**.

Stage 1 — CBP Custody

This represents the population currently in CBP custody.

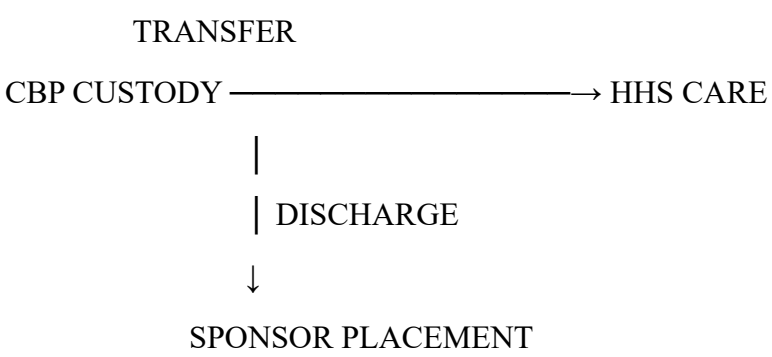
Stage 2 — HHS Care

Children transferred from CBP enter HHS care, where services such as medical screening, sheltering, and case management take place.

Stage 3 — Sponsor Placement

Children discharged from HHS care represent the successful exit from the observed care pipeline through sponsor placement.

The analytical model is:



An important analytical distinction is that **CBP custody and HHS care are stock measures**, whereas **transfers and discharges are flow measures**. Therefore, ratios involving these variables should be interpreted as operational performance indicators rather than individual-level transition probabilities.

7. Data Preparation

The dataset was prepared for analysis through the following steps:

7.1 Removal of Empty Records

Completely blank records were removed from the dataset.

7.2 Date Conversion

The reporting date was converted to a datetime format to support:

- monthly analysis,
- weekday/weekend analysis,
- time-series visualization,
- trend analysis.

7.3 Numeric Conversion

Numeric fields were converted into appropriate numerical data types.

The HHS Care variable required removal of comma separators before numerical conversion.

7.4 Column Standardization

For analytical convenience, the original columns were renamed:

date

apprehended

cbp_custody

transferred

hhs_care

discharged

7.5 Derived Variables

The following analytical variables were created:

transfer_efficiency

discharge_effectiveness

net_flow

flow_status

month

day_type

8. Exploratory Data Analysis

A. Care Pipeline Modeling

The exploratory analysis examines the volume and behaviour of the major stages of the care pipeline.

The analysis focuses on:

- CBP custody levels
- HHS care levels
- Transfer volumes
- Discharge volumes
- Inflow versus successful exits
- Temporal trends
- Variability in placement activity

The EDA establishes the foundation for the transition efficiency and bottleneck analysis.

- **CBP custody levels:**

  CBP custody levels refer to the number of unaccompanied children who are currently in U.S. Customs and Border Protection (CBP) custody at a given reporting point.

CBP Custody	Date
53	2023-01-12
49	2023-01-22
50	2023-01-23
42	2023-01-24
22	2023-01-25

  For your **Care Transition Efficiency & Placement Outcome Analytics** project, CBP custody levels help us assess the **first stage of the care transition pipeline**:

Apprehension → CBP Custody → HHS Transfer → HHS Care → Discharge

A persistently high CBP custody level may indicate that children are remaining in the initial custody stage while awaiting transfer, although **CBP custody alone cannot establish a delay**.

HHS Care Levels

HHS care levels represent the number of unaccompanied children who are **currently in HHS care at a given reporting point**.

It is a **stock measure**, meaning it reflects the active number of children in HHS care rather than the number entering or leaving HHS during the reporting period.

Date	HHS Care Level
2023-01-12	6,566
2023-01-22	7,122
2023-01-23	7,280
2023-01-24	7,433
2023-01-25	7,538

The HHS care level increased across these reporting observations, indicating a growing number of children recorded within the HHS care stage during this period.

- **Transfer volumes**

Transfer volumes represent the number of unaccompanied children transferred from CBP custody to HHS care during a reporting period.

Date	Transfer Volume
2023-01-12	34
2023-01-22	39
2023-01-23	39
2023-01-24	47
2023-01-25	41

he transfers volume therefore indicates the movement of children from the **CBP stage to the HHS stage**.

Why transfer volumes are important

Transfer volumes help evaluate the **CBP → HHS transition efficiency**. They can be analysed alongside:

- cbp_custody — children currently in CBP custody
- apprehended — new children entering CBP custody
- hhs_care — children currently in HHS care

A sustained reduction in transfer volumes, particularly when CBP custody is elevated, may warrant further investigation as a potential **transition bottleneck**. However, transfer volume alone does not establish that a delay occurred.

- **Discharge volumes**

Discharges from HHS care range from 175 to 436 in the sample. The highest discharge volume is observed on January 12, 2023, with 436 discharges.

Care Pipeline Modeling

The Care Pipeline Modeling framework represents the UAC process as a **multi-stage flow pipeline**, allowing the movement of children through the major stages of care to be analyzed systematically. Based on the project framework, the pipeline is represented as:

CBP Custody → HHS Care → Sponsor Placement

The analysis distinguishes between **stock measures**, which represent children present at a reporting point, and **flow measures**, which represent movements during the reporting period.

1. Representation of the System as a Flow Pipeline

The care transition process was modeled using five key variables from the dataset:

Pipeline Component	Dataset Variable	Type	Description
New intake	apprehended	Flow	Children apprehended and placed in CBP custody
CBP stage	cbp_custody	Stock	Children recorded in CBP custody
CBP → HHS movement	Transferred	Flow	Children transferred from CBP custody to HHS
HHS stage	hhs_care	Stock	Children recorded in HHS care
HHS → Sponsor placement	Discharged	Flow	Children discharged from HHS care

This structure enables the analysis to move beyond simply measuring the number of children in custody and instead examine **how children move through the overall care pipeline**.

2. Definition of Pipeline Stages

The first stage is **CBP custody**, where the dataset records the number of children in CBP custody at the reporting point. In the sample observations, CBP custody ranges from **22 to 53 children**.

The second stage is **HHS care**. Children transferred from CBP enter this stage, while children subsequently leave through discharge. In the sample data, HHS care levels increase from **6,566 on January 12, 2023, to 7,538 on January 25, 2023**.

The final stage is **Sponsor Placement**, represented in the dataset by the discharged variable. Discharge volumes indicate successful exits from HHS care. In the sample observations, discharge volume ranges from **175 to 436 children**, with the highest value of 436 recorded on January 12, 2023.

3. Tracking Movements Between Stages

Daily or reporting-period movements were tracked using the transferred and discharged variables.

For the displayed observations:

- **January 12, 2023:** 34 children were transferred to HHS and 436 were discharged.
- **January 22, 2023:** 39 children were transferred and 227 were discharged.
- **January 23, 2023:** 39 children were transferred and 181 were discharged.
- **January 24, 2023:** 47 children were transferred and 175 were discharged.
- **January 25, 2023:** 41 children were transferred and 180 were discharged.

A useful indicator for examining movement within the HHS stage is:

$$\text{Net HHS Movement} = \text{Transfers} - \text{Discharges}$$

For example, on **January 24, 2023**:

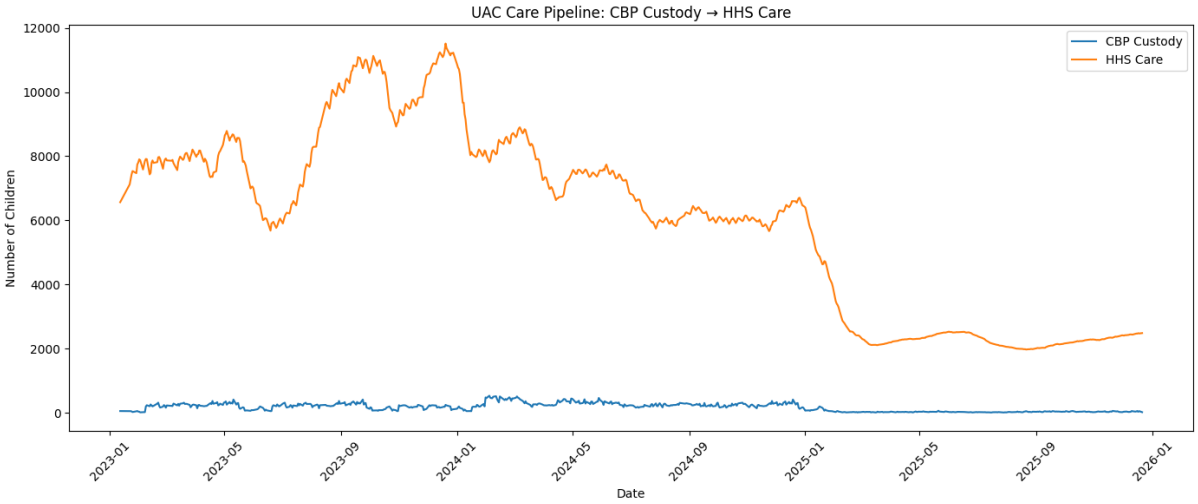
$$47 - 175 = -128$$

This indicates that **discharges exceeded transfers by 128 children** during that reporting observation. Conversely, when transfers exceed discharges, the net movement is positive and indicates greater inflow than successful exits for that observation.

4. Pipeline Modeling Interpretation

The pipeline model therefore provides a structured view of the relationship between **incoming cases, active care populations, transfers, and successful exits**. Rather than evaluating each measure independently, the analysis considers how movements between stages affect the overall pipeline.

The sample data demonstrates that the relationship between transfers and discharges varies across reporting observations. At the same time, the HHS care population changes substantially, increasing from **6,566 to 7,538** across the displayed observations. These patterns justify further analysis of sustained inflow–exit imbalances and potential accumulation periods using the complete dataset.



Care Pipeline Modeling – CBP Custody and HHS Care Trends

The **Care Pipeline Modeling** analysis represents the UAC process as a sequential flow:

CBP Custody → HHS Care → Sponsor Placement

The chart above compares the number of children recorded in **CBP custody** and **HHS care** across the reporting period from approximately **January 2023 to December 2025**. These two measures are treated as **stock measures**, representing the number of children in each stage at a reporting point.

1. CBP Custody Levels

The blue line represents children in CBP custody. Compared with HHS care, CBP custody remains at a substantially lower level throughout the period.

The chart shows:

- CBP custody generally remains in the **low hundreds**.
- There are periods of increased CBP custody, particularly around **late 2023 and early 2024**.
- A noticeable reduction occurs around **early 2025**, after which CBP custody remains at relatively low levels.
- The fluctuations indicate changes in the number of children present in the CBP stage at different reporting points.

2. HHS Care Levels

The orange line represents the number of children in HHS care and shows considerably greater volume and variation.

Key patterns include:

- HHS care levels begin at approximately **6,500–7,000** children in early 2023.
- Levels increase during 2023, reaching approximately **11,000–11,500** toward late 2023.

- Another elevated period occurs around late 2023 to early 2024.
- From early 2024 onward, HHS care levels show a generally declining pattern, with fluctuations.
- During late 2024, levels remain around **6,000–7,000**.
- A **sharp decline occurs around early 2025**, with HHS care falling to approximately **2,000–3,000**.
- Following this decline, HHS care levels remain relatively low and show a modest increase toward the end of 2025.

3. Relationship Between CBP Custody and HHS Care

The chart demonstrates a substantial difference in the scale of the two pipeline stages. HHS care consistently contains a much larger recorded population than CBP custody.

This difference is important for the pipeline analysis because it indicates that the **HHS stage represents the larger active care population** within the observed system. Changes in HHS care levels therefore provide an important indication of changes in the overall care workload.

However, the chart alone does **not establish the presence or absence of a backlog**. To identify accumulation, HHS care levels need to be analyzed together with **transfers into HHS and discharges from HHS**.

4. Overall Pipeline Interpretation

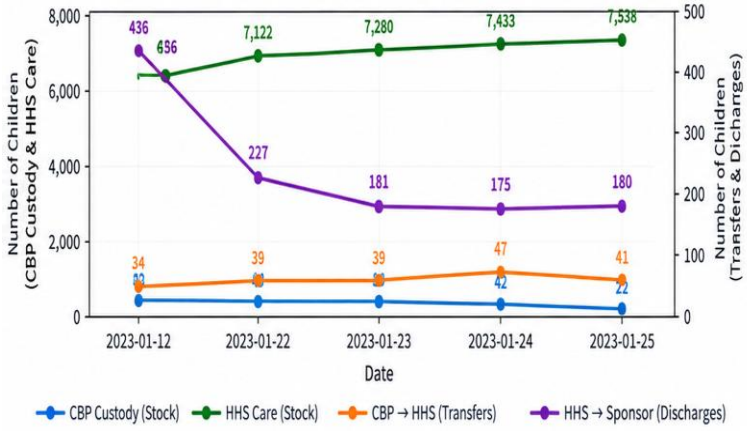
The observed pattern suggests three broad phases:

1. 2023 – High and increasing HHS care levels:
   HHS care rises from approximately 6,500 to above 10,000, with the highest levels occurring toward the end of the year.
2. 2024 – Gradual reduction with fluctuations:
   HHS care decreases from its earlier peak but continues to fluctuate between approximately 6,000 and 9,000 for much of the year.
3. 2025 – Sharp reduction and lower-volume period:
   HHS care experiences a pronounced decline around the beginning of 2025 and subsequently remains around 2,000–2,500 for much of the later period shown.

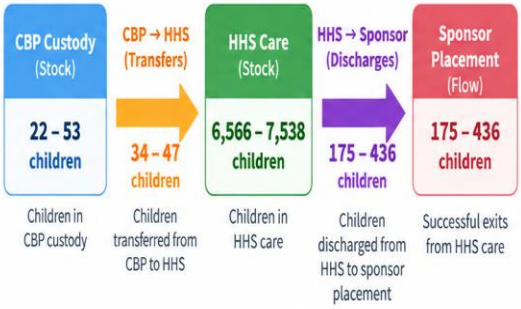
UAC Care Pipeline - Sample Data

Date	CBP Custody (Stock)	CBP → HHS (Transfers)	HHS Care (Stock)	HHS → Sponsor (Discharges)	Discharged (Flow)
2023-01-12	53	34	6,566	436	436
2023-01-22	49	39	7,122	227	227
2023-01-23	50	39	7,280	181	181
2023-01-24	42	47	7,433	175	175
2023-01-25	22	41	7,538	180	180

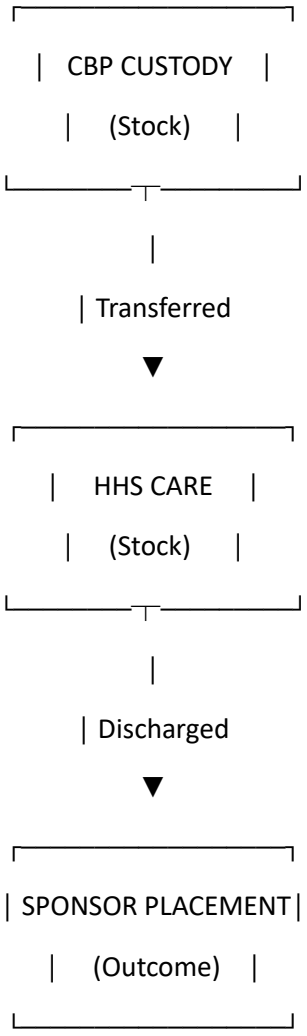
UAC Care Pipeline Trends (Sample Period)



UAC Care Pipeline Flow



UAC CARE PIPELINE



Stage 1 – CBP Custody

The first stage represents children recorded in **CBP custody**. The variable cbp_custody is treated as a **stock measure**, indicating the number of children in CBP custody at a reporting point.

Stage 2 – HHS Care

Children transferred from CBP custody move into the **HHS Care** stage. The transferred variable represents the movement from CBP to HHS, while hhs_care represents the active population recorded in HHS care.

Stage 3 – Sponsor Placement

The final stage represents **Sponsor Placement**. In the dataset, this stage is represented through **discharges from HHS care**. The discharged variable therefore represents successful exits from the HHS care stage to sponsor placement.

Pipeline Movement

The two key transitions are:

$$\text{CBP Custody} \xrightarrow{\text{Transfers}} \text{HHS Care}$$

and

$$\text{HHS Care} \xrightarrow{\text{Discharges}} \text{Sponsor Placement}$$

For the sample observations provided:

- **CBP → HHS transfers:** 34–47 children
- **HHS → Sponsor discharges:** 175–436 children
- **CBP custody:** 22–53 children
- **HHS care:** 6,566–7,538 children

B. Transition Efficiency Metrics

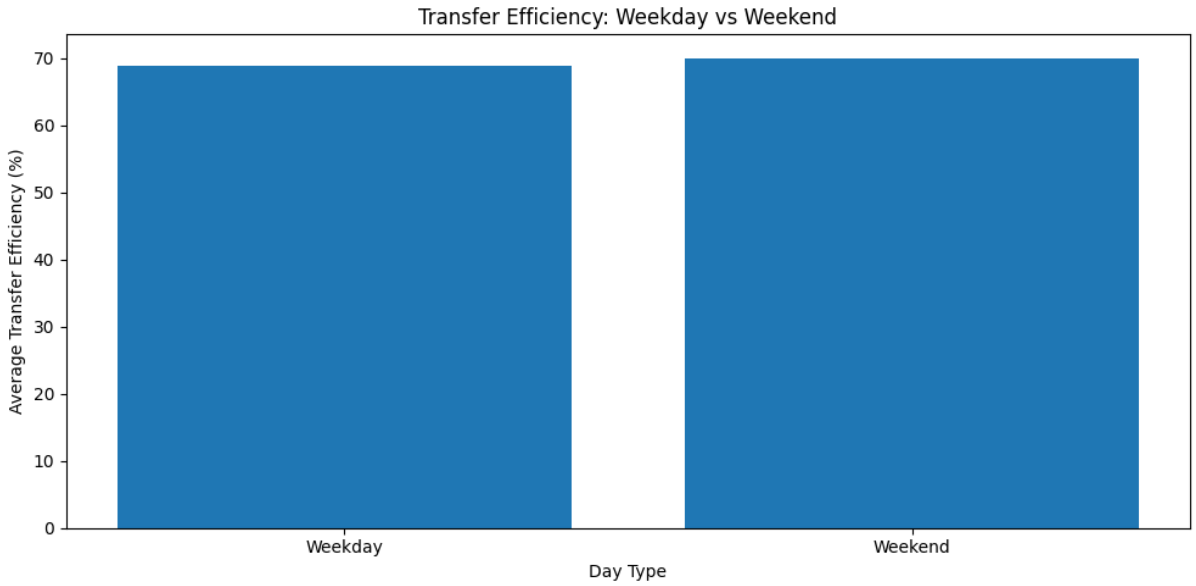
Three core process metrics are used.

1. Transfer Efficiency Ratio

The Transfer Efficiency Ratio is defined as:

$$Transfer\ Efficiency\ Ratio = \frac{Transfers}{CBP\ Custody}$$

This metric provides an operational indicator of transfer activity relative to the CBP custody population.



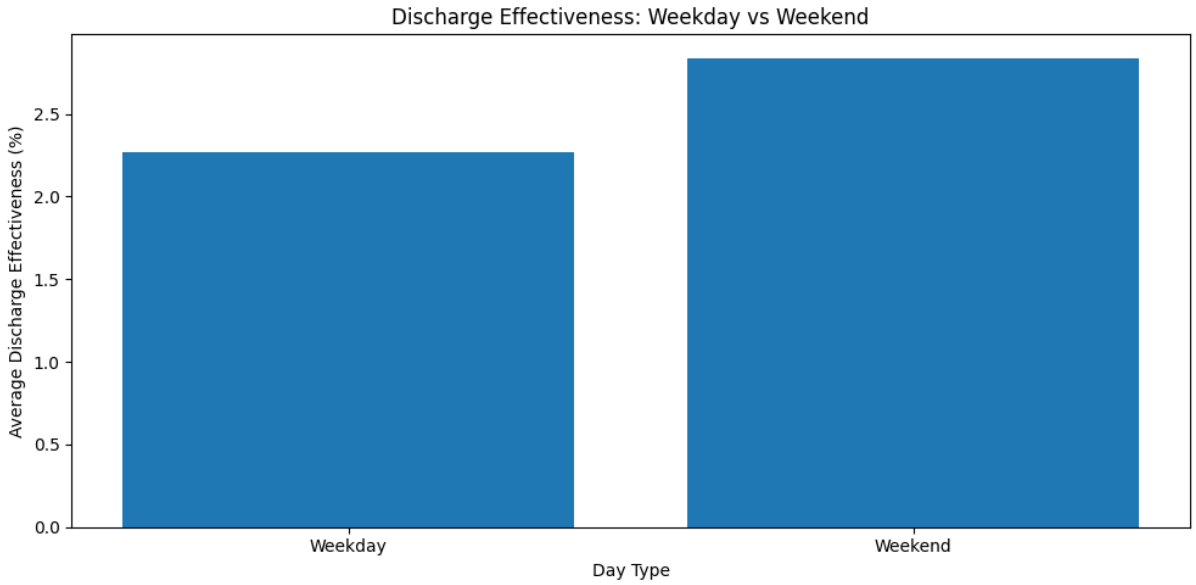
Transfer efficiency was compared across weekday and weekend reporting observations to identify potential differences in CBP-to-HHS transition performance. The average transfer efficiency was approximately **68.8% on weekdays** and **70.0% on weekends**. The difference of approximately **1.2 percentage points** indicates relatively similar transfer efficiency across the two day types, with weekend observations showing a slightly higher average. Therefore, the analysis does not indicate a substantial weekday–weekend difference in transfer efficiency based on the available observations. Because the dataset contains gaps in reporting dates, the comparison should be interpreted as a comparison of available weekday and weekend reporting observations rather than continuous daily operations.

2. Discharge Effectiveness

Discharge Effectiveness is defined as:

$$Discharge\ Effectiveness = \frac{Discharges}{HHS\ Care}$$

This measures discharge activity relative to the HHS care population.



Weekday vs. Weekend Transfer Efficiency:

Transfer efficiency was compared across weekday and weekend reporting observations to identify potential differences in CBP-to-HHS transition performance. The average transfer efficiency was approximately **68.8% on weekdays** and **70.0% on weekends**. The difference of approximately **1.2 percentage points** indicates relatively similar transfer efficiency across the two day types, with weekend observations showing a slightly higher average. Therefore, the analysis does not indicate a substantial weekday–weekend difference in transfer efficiency based on the available observations. Because the dataset contains gaps in reporting dates, the comparison should be interpreted as a comparison of available weekday and weekend reporting observations rather than continuous daily operations.

3. Pipeline Throughput Rate

Pipeline Throughput Rate is defined as:

$$Pipeline\ Throughput\ Rate = \frac{Total\ Exits}{Total\ Entries}$$

For this project:

Total Entries = Children apprehended and placed in CBP custody

Total Exits = Children discharged from HHS care

These three metrics form the core transition-efficiency framework specified in the project methodology.

UAC Care Pipeline - Transition Efficiency Metrics



1. Transfer Efficiency Ratio

Transfers ÷ CBP Custody

0.75
(75.03%)

Indicates the proportion of children in CBP custody who are transferred to HHS care.

2. Discharge Effectiveness

Discharges ÷ HHS Care

0.03
(2.86%)

Indicates the proportion of children in HHS care who are discharged to sponsor placement.

3. Pipeline Throughput Rate

Total Exits ÷ Total Entries

1.85
(185.42%)

Indicates the overall throughput of the pipeline, comparing total exits to total entries.

Metric	Formula	Value	Percentage	Interpretation
Transfer Efficiency Ratio	Transfers ÷ CBP Custody	0.75	75.03%	75.03% of children in CBP custody were transferred to HHS care.
Discharge Effectiveness	Discharges ÷ HHS Care	0.03	2.86%	2.86% of children in HHS care were discharged to sponsor placement.
Pipeline Throughput Rate	Total Exits ÷ Total Entries	1.85	185.42%	Overall pipeline throughput rate of 185.42%.

C. Backlog and Delay Identification

Backlog analysis compares incoming transfers with successful HHS discharges.

The principal measure is:

$$Net\ Flow = Transfers - Discharges$$

Interpretation

Net Flow	Interpretation
Positive	Inflow exceeds exits; accumulation pressure
Negative	Exits exceed inflow; potential backlog reduction
Zero	Inflow and exits are balanced

The analysis identifies sustained periods in which transfers remain above discharges.

A cumulative accumulation indicator is also calculated:

$$Cumulative\ Accumulation = \sum (Transfers - Discharges)$$

This is treated as an **accumulation-pressure indicator**, rather than a literal count of unresolved individual cases.

1. Inflow vs. Successful Exits

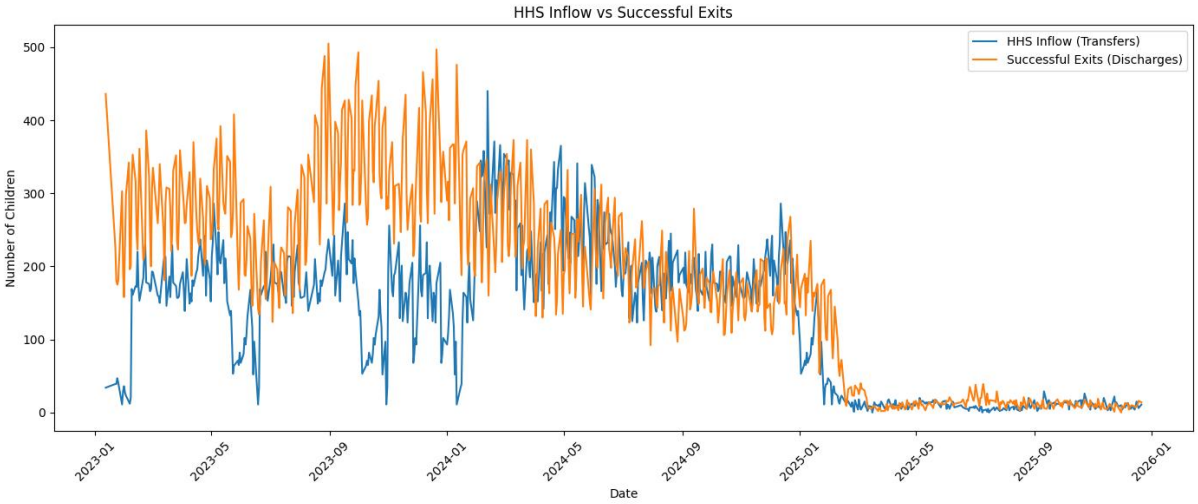
The following metric was used to assess the balance between incoming and outgoing cases:

$$\text{Net Flow} = \text{Successful Exits} - \text{Inflow}$$

where:

- **Inflow** = transferred
- **Successful Exits** = discharged

A **negative net flow** indicates that exits exceeded inflow, while a **positive net flow** indicates that inflow exceeded exits and may represent potential accumulation pressure.



HHS Inflow and Successful Exit Analysis:

The comparison of HHS inflows and successful exits shows substantial variability during 2023 and early 2024. During this period, transfers into HHS care and discharges from HHS care frequently fluctuated between approximately 100 and 500 children per reporting observation, with several periods in which successful exits exceeded incoming transfers. During mid-to-late 2024, the two measures generally moved closer together while their overall volumes declined. A pronounced reduction in both inflows and successful exits occurred around early 2025, followed by a sustained period of relatively low-volume activity. The visual comparison provides an overview of changes in pipeline activity; however, potential backlog accumulation was assessed separately using net flow and sustained imbalance measures rather than relying on the chart alone.

2. Results from the Sample Period

The sample observations show the following:

Date	Inflow	Successful Exits	Net Flow	Status
2023-01-12	34	436	-402	Exits > Inflow
2023-01-22	39	227	-188	Exits > Inflow
2023-01-23	39	181	-142	Exits > Inflow

Date	Inflow	Successful Exits	Net Flow	Status
2023-01-24	47	175	-128	Exits > Inflow
2023-01-25	41	180	-139	Exits > Inflow
2023-01-29	11	303	-292	Exits > Inflow
2023-01-30	29	196	-167	Exits > Inflow
2023-01-31	36	158	-122	Exits > Inflow
2023-02-01	27	231	-204	Exits > Inflow
2023-02-02	23	298	-275	Exits > Inflow

Across all **10 displayed observations**, successful exits were greater than incoming transfers.

The net flow ranged from **-402 to -122 children**. The largest negative net flow occurred on **January 12, 2023**, when 436 children were discharged compared with 34 transfers.

3. Sustained Imbalance

The sample demonstrates a sustained pattern in which:

Successful exits > HHS inflow

throughout the displayed period.

For example:

- January 12: 436 exits vs. 34 inflows
- January 24: 175 exits vs. 47 inflows
- February 2: 298 exits vs. 23 inflows

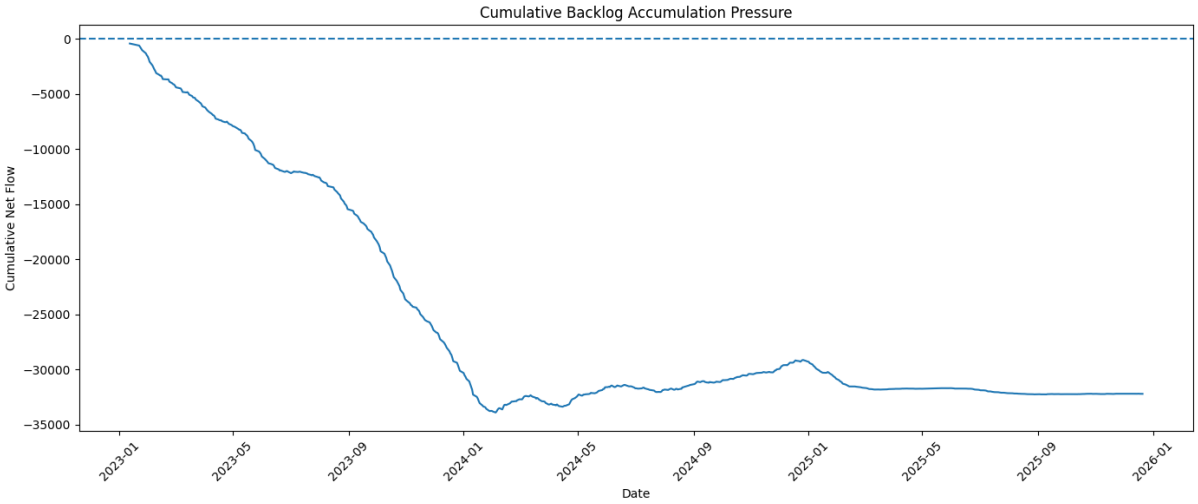
This indicates that, within these observations, the number of children exiting HHS care was greater than the number entering through transfers.

4. Accumulation Assessment

Because inflow did not exceed successful exits in any of the displayed observations, the sample does **not show an inflow-driven accumulation pattern** during January 12–February 2, 2023.

The negative net-flow values suggest that the balance of transfers and discharges was associated with a **reduction rather than an increase in the active HHS flow burden** during these observations.

However, this should not be interpreted as proof that there was no backlog anywhere in the system. The analysis only evaluates the relationship between **transfers and discharges**. In addition, the dataset contains gaps between reporting dates, so these observations should not be interpreted as continuous daily measurements.



Cumulative Backlog Accumulation Pressure

The cumulative backlog analysis was performed to examine the **long-term balance between HHS inflows and successful exits**. The purpose of this analysis was to identify whether the care pipeline showed a sustained tendency toward accumulation or reduction over the reporting period.

The cumulative measure was calculated as:

$$\text{Net Flow} = \text{Inflow} - \text{Successful Exits}$$

where:

- **Inflow** = children transferred into HHS care
- **Successful Exits** = children discharged from HHS care

The cumulative value was then calculated as:

$$\text{Cumulative Net Flow}_t = \sum_{i=1}^{t} (\text{Inflow}_i - \text{Successful Exits}_i)$$

1. Overall Trend

The chart shows a **strong downward trend in cumulative net flow**, beginning near zero in early 2023 and reaching approximately **−34,000** by early 2024.

This indicates that, across the accumulated reporting observations, **successful exits were substantially greater than incoming transfers**.

2. 2023 – Strong Reduction in Cumulative Flow

During 2023, the cumulative net flow declines consistently.

- The measure moves from approximately **0** at the beginning of the period.
- It falls below **−10,000** by approximately mid-2023.

- The decline becomes steeper during the second half of 2023.
- By the beginning of 2024, cumulative net flow reaches approximately **−33,000 to −34,000**.

This pattern indicates a sustained period in which recorded HHS exits exceeded recorded transfers.

3. 2024 – Stabilization

During much of 2024, the cumulative measure becomes relatively stable.

After reaching approximately **−34,000**, the line rises moderately toward approximately **−31,000 to −32,000** and then remains within a relatively narrow range.

This suggests that the difference between inflows and exits became more balanced compared with the substantial negative accumulation observed during 2023.

4. 2025 – Further Stabilization

Around early 2025, the cumulative measure shows another decline, reaching approximately **−32,000** and subsequently remaining relatively stable through the remainder of the reporting period.

The relatively flat pattern indicates that the cumulative difference between transfers and discharges changed less substantially during this period.

5. Interpretation for Backlog Analysis

The negative cumulative values should **not be interpreted as a literal negative number of children or as a direct count of resolved cases**.

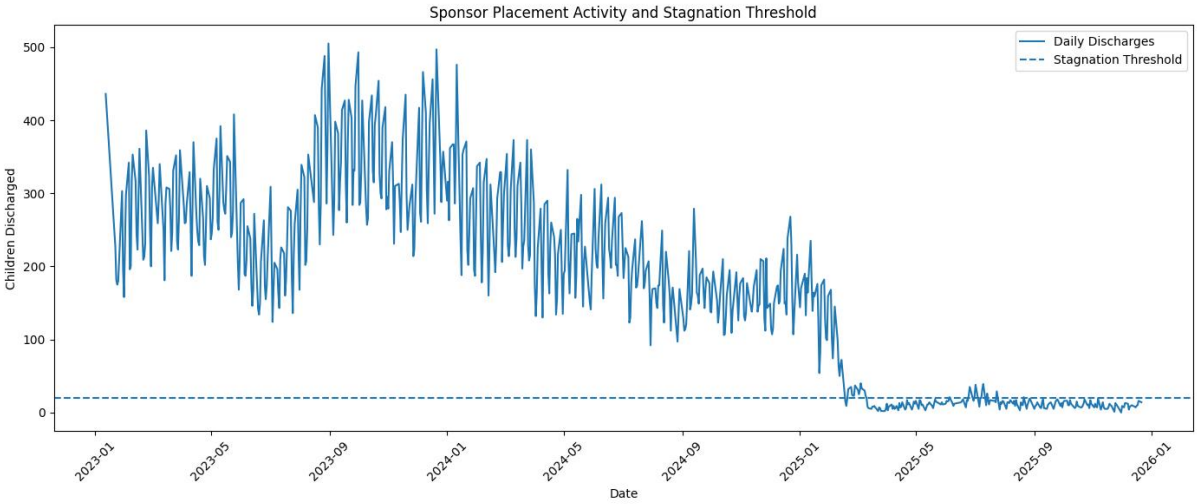
Instead, they indicate that, based on the selected flow definition:

Cumulative HHS exits exceeded cumulative HHS inflows.

Therefore, the chart does **not show evidence of sustained inflow-driven accumulation** across the observed reporting period. Instead, it shows a cumulative reduction in the transfer-versus-discharge flow balance.

Important Analytical Limitation

Because the dataset contains **gaps between reporting dates**, the cumulative ca



Sponsor Placement Activity and Stagnation Analysis

The **Sponsor Placement Activity and Stagnation Threshold** analysis examines the number of children discharged from HHS care over the reporting period and identifies observations where discharge activity falls below a predefined stagnation threshold.

The discharged variable represents the number of children exiting HHS care through sponsor placement. The threshold shown by the dashed horizontal line is based on the **25th percentile of discharge volumes**, which provides a data-driven reference point for relatively low discharge activity.

1. Overall Discharge Pattern

The chart shows substantial variation in discharge volumes between **January 2023 and December 2025**.

- During **2023**, discharge volumes are generally high and highly variable, with many observations between approximately **150 and 400** children and several peaks approaching **500**.
- Around **late 2023 and early 2024**, discharge activity remains relatively high, although variability continues.
- During **2024**, discharge volumes show a gradual downward pattern, with most observations falling approximately between **100 and 300**.
- A pronounced reduction occurs around **early 2025**, when discharge volumes fall sharply.
- After this reduction, discharge activity remains substantially lower, generally close to the **10–30 children** range.

2. Stagnation Threshold

The dashed horizontal line represents the **stagnation threshold**, calculated using the 25th percentile of discharge volume.

This threshold is used as an analytical flag rather than as a fixed operational definition of failure.

Observations below the threshold indicate periods of **relatively low discharge activity** compared with the overall distribution of discharge volumes.

3. Identification of Low-Activity Periods

The chart indicates that most of the reporting observations during **2023 and 2024** remain above the stagnation threshold.

The concentration of observations at or below the threshold occurs primarily after the sharp decline in **early 2025**. This indicates a prolonged period of comparatively low sponsor-placement activity.

It is important to distinguish **low discharge activity** from a confirmed backlog. A low number of discharges could result from lower inflows or other changes in the system. Therefore, stagnation should be evaluated alongside HHS care levels, transfers, and inflow–exit balance.

D. Temporal and Pattern Analysis

The temporal analysis examines three areas.

1. Weekday vs Weekend

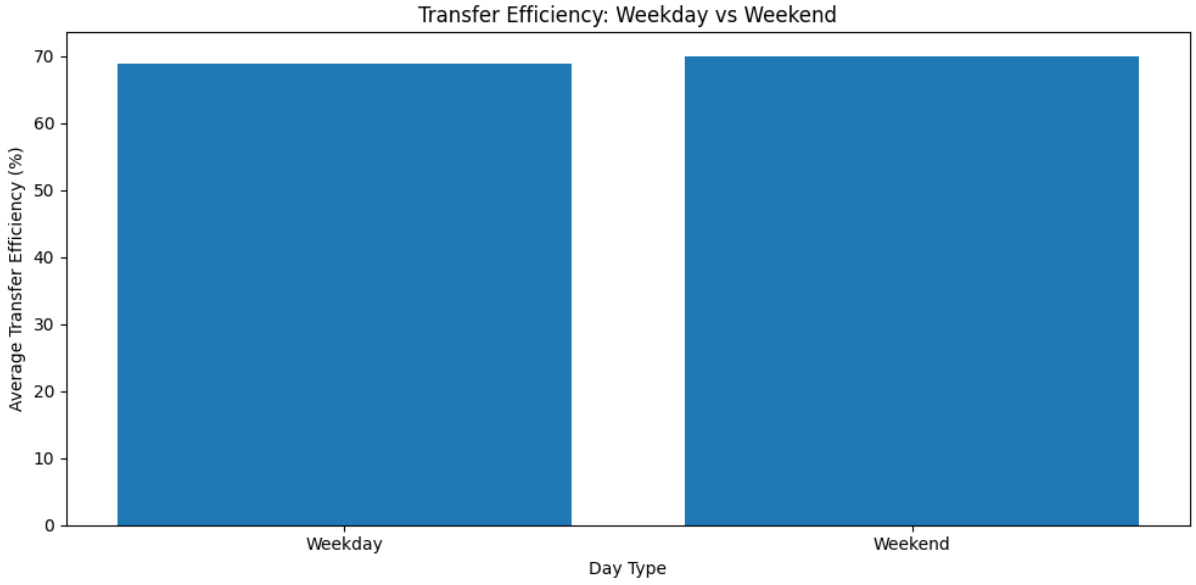
Reporting observations are categorized into:

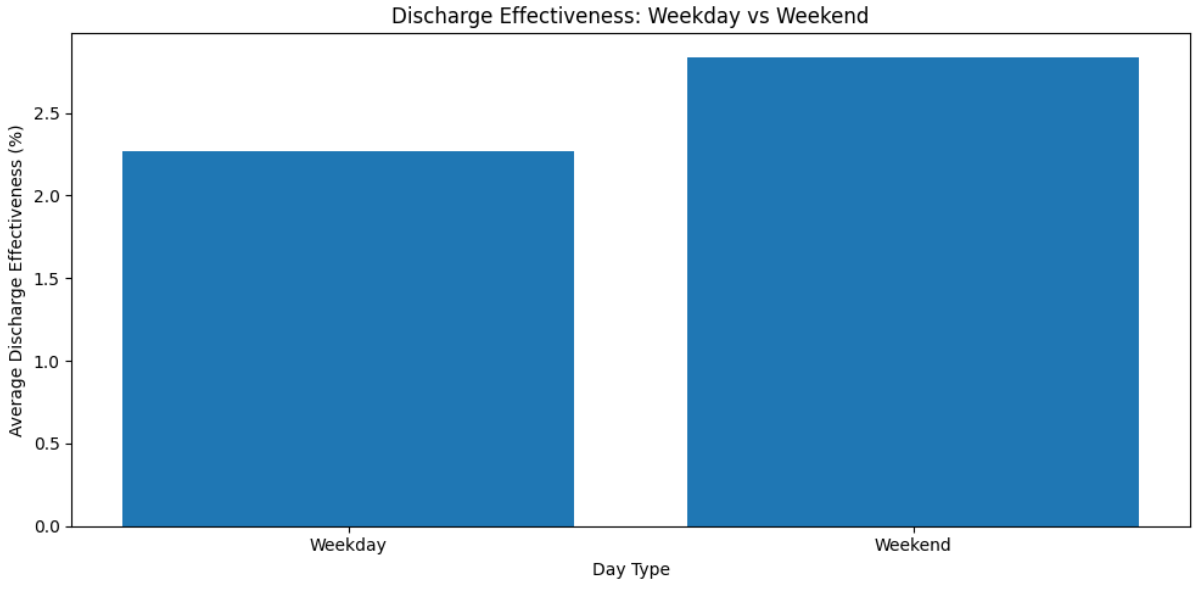
Weekday

Weekend

Transfer efficiency and discharge effectiveness are then compared between the two groups.

Because the dataset contains reporting-date observations rather than individual case-level timestamps, this analysis describes performance across observed reporting dates.





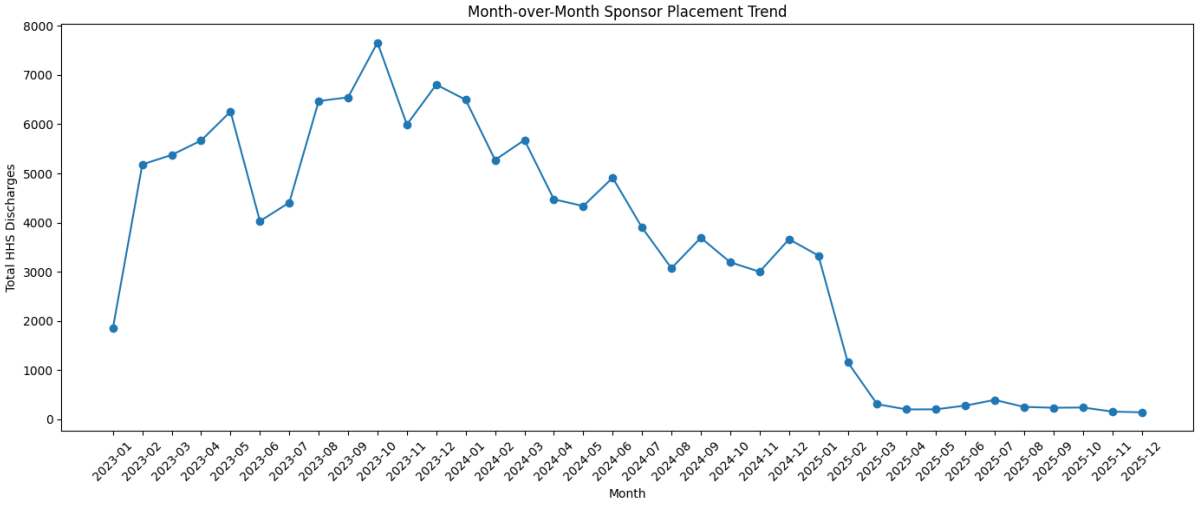
11.2 Month-over-Month Placement Trends

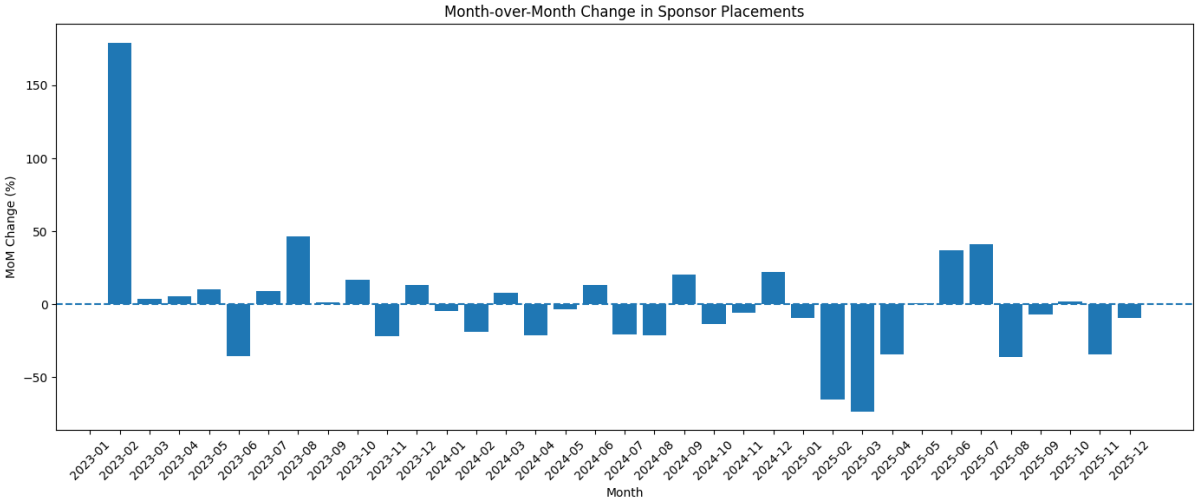
Monthly discharge totals are calculated to identify:

- increases in placement activity,
- decreases in placement activity,
- periods of stability,
- major changes in monthly discharge volume.

The month-over-month change is calculated as:

$$MoM\ Change = \frac{Current\ Month - Previous\ Month}{Previous\ Month} \times 100$$





3. Prolonged Stagnation

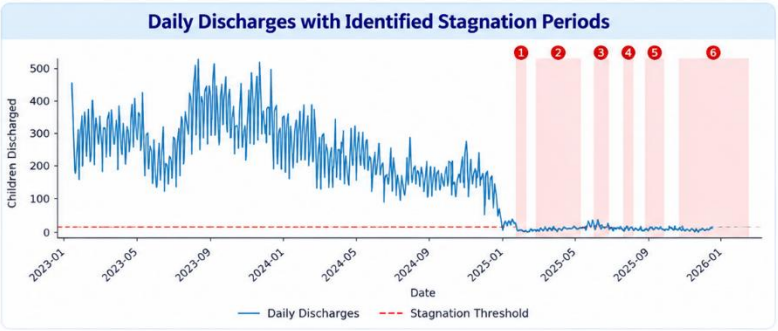
Low-discharge observations are identified using a data-driven threshold.

The analysis groups consecutive low-discharge reporting observations to identify periods of potential

Stagnation

Identified Stagnation Periods (Low Sponsor Placement Activity)

	Start Date	End Date	Observations (Days)	Total Discharges	Average Discharges (per day)
1	2025-02-17	2025-02-19	3	43	14.33
2	2025-03-12	2025-06-04	59	544	9.22
3	2025-06-08	2025-06-24	8	106	13.25
4	2025-07-15	2025-07-23	6	91	15.17
5	2025-07-27	2025-08-20	19	227	11.95
6	2025-08-24	2025-12-21	82	849	10.35



Key Insights

- Six stagnation periods were identified in 2025 based on discharge activity falling below the defined threshold.
- The longest stagnation period was from 2025-08-24 to 2025-12-21, with 82 observations and an average of 10.35 discharges per day.
- The lowest average daily discharges occurred during 2025-03-12 to 2025-06-04, with 9.22 discharges per day.
- All identified periods show significantly lower discharge volumes compared to early 2023 and 2024, when daily discharges were frequently between 100 and 400.
- These periods indicate relatively low sponsor-placement activity and should be evaluated alongside inflow and HHS care levels to assess potential operational impacts.

These periods are interpreted as **prolonged low placement activity**, rather than proof that individual children remained unresolved.

E. Outcome Stability Analysis

Outcome stability was evaluated using **discharge volumes** as the indicator of placement outcomes. The analysis focused on three areas:

1. **Variability in discharge performance**
2. **Consistency of placement outcomes**
3. **Identification of potential sudden drops in discharge activity**

The dataset contains **720 reporting observations** of discharge activity.

1. Variability in Discharge Performance

Descriptive statistics were calculated for the discharged variable:

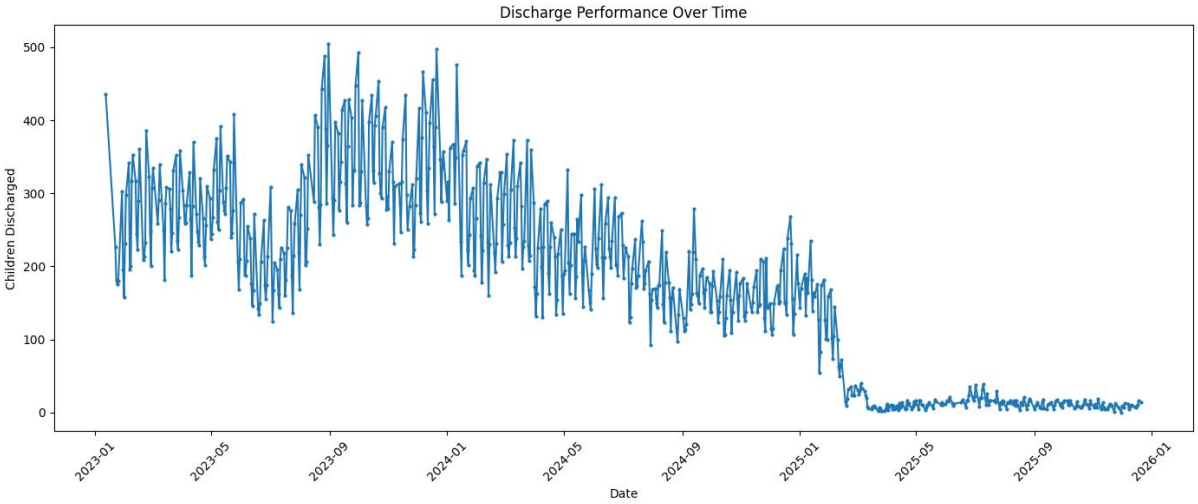
Statistic	**Value**
Observations	720
Mean	173.41
Standard deviation	125.70
Minimum	0.00
25th percentile	19.75
Median	181.00
75th percentile	267.00
Maximum	505.00

The average discharge volume was **173.41 children per reporting observation**, with a standard deviation of **125.70**. The relatively large difference between the mean and standard deviation indicates considerable variation in discharge activity across the reporting period.

The discharge volume ranged from **0 to 505 children**, demonstrating substantial differences in placement activity between reporting observations.

The median discharge volume was **181 children**, while the 25th percentile was only **19.75 children**. This large difference indicates that a substantial portion of the observations occurred at much lower discharge levels, particularly toward the later part of the dataset.

Discharge Performance Over Time



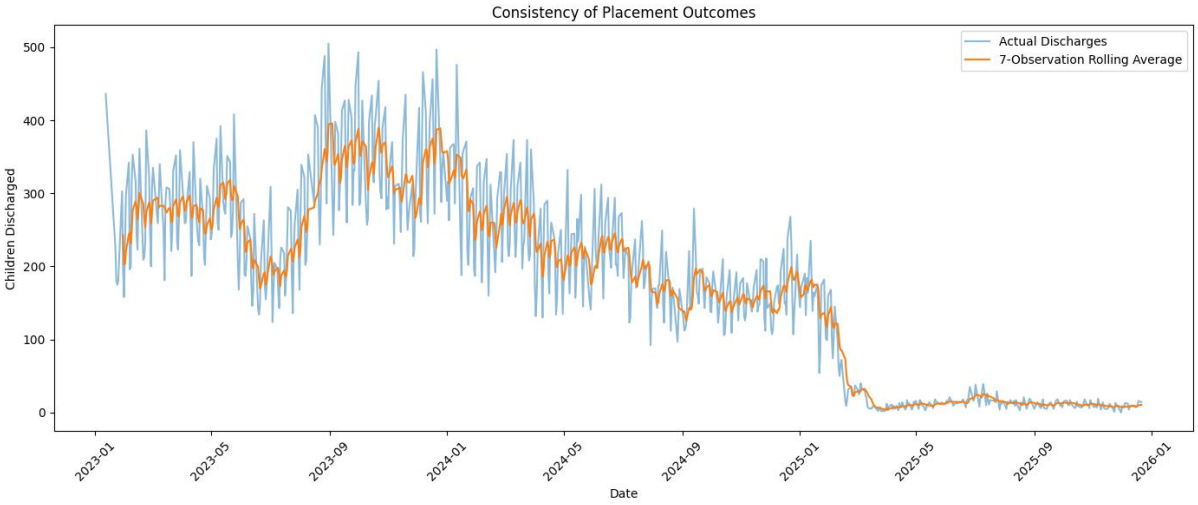
The discharge performance trend shows considerable variation during **2023**. Discharge volumes frequently ranged between approximately **150 and 400 children**, with several observations exceeding **400 children** and reaching a maximum of **505 children**.

During **2024**, the overall level of discharge activity gradually declined. Although substantial fluctuations remained, the typical discharge volume became lower compared with the earlier period.

A pronounced reduction is visible around **early 2025**. Following this decline, discharge volumes remained at substantially lower levels for the remainder of the reporting period.

This represents an important change in the observed placement activity and requires consideration alongside other pipeline indicators such as HHS care levels and transfer volumes.

2. Consistency of Placement Outcomes



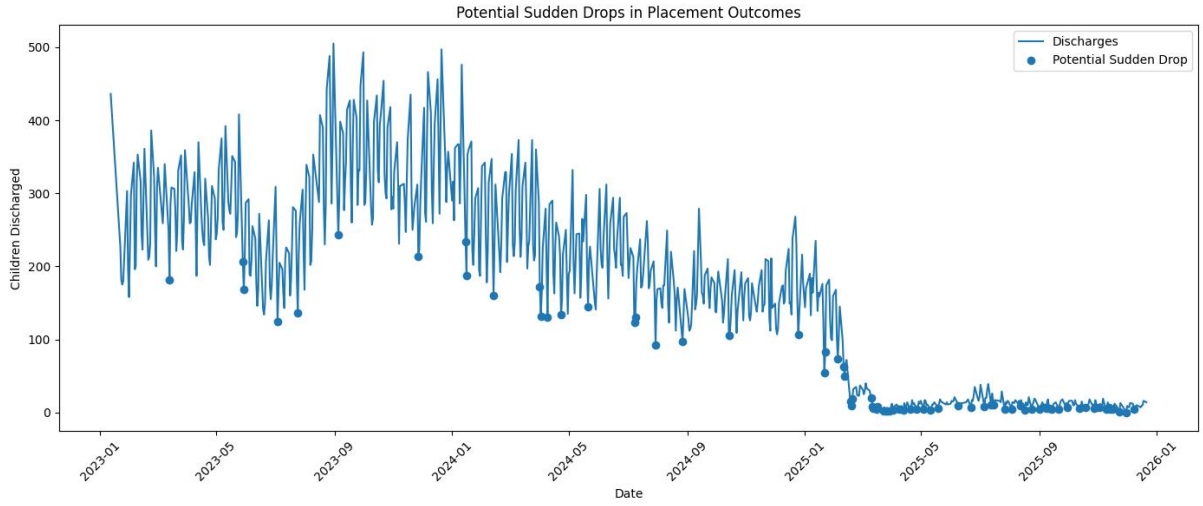
A **7-observation rolling average** was used to smooth short-term fluctuations and examine the underlying discharge pattern.

The rolling-average analysis shows that:

- Discharge activity was relatively high during much of 2023.
- The rolling average reached its highest levels around late 2023 and early 2024.
- The rolling average gradually declined during 2024.
- A sharp reduction occurred around early 2025.
- After the reduction, the rolling average remained at a substantially lower level.

The rolling average therefore confirms that the reduction observed in 2025 was not simply caused by one isolated low-discharge observation. Instead, the data show a **persistent change in the level of discharge activity**.

3. Potential Sudden Drops



Potential sudden drops were identified by comparing individual discharge observations with the preceding discharge level and flagging observations showing a substantial decline.

The analysis identifies multiple potential sudden-drop observations throughout the dataset. These occur during both periods of relatively high discharge activity and the later low-volume period.

The concentration of low discharge observations after early 2025 is particularly notable because the decline is both **large and persistent** rather than limited to a single observation.

However, a statistical drop in discharge volume does not by itself establish the reason for the change. Further investigation would require comparing discharge activity with **HHS care levels, transfer volumes, and inflows**.

- **Outcome Stability Interpretation**

The results indicate that discharge performance was **highly variable across the full reporting period**. The mean of 173.41 and standard deviation of 125.70 demonstrate substantial

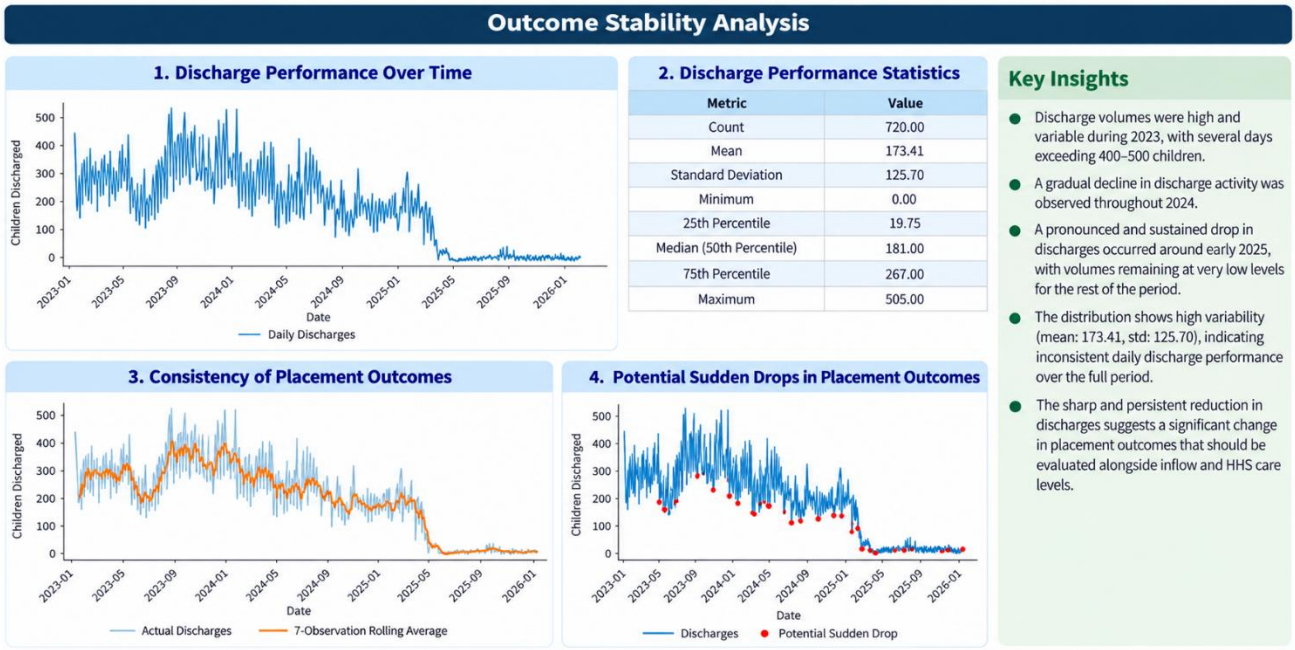
variation, while the range from 0 to 505 further illustrates the wide spread of observed discharge volumes.

The time-series and rolling-average analyses additionally show a **structural change in discharge activity around early 2025**, followed by a prolonged period of much lower discharge volumes.

Because the dataset does not contain a separate variable explicitly measuring "reunification success," this analysis uses **discharges as the available indicator of successful exits/sponsor placement**, consistent with the project framework. Therefore, the results should be described as **discharge or placement outcome stability**, rather than directly as individual-level reunification success

- **Outcome Stability Analysis:**

Outcome stability was assessed using discharge volumes as an indicator of placement activity. Across 720 reporting observations, the mean discharge volume was **173.41 children**, with a standard deviation of **125.70**. Discharge volumes ranged from **0 to 505**, while the median was **181 children**. The time-series analysis showed high and variable discharge activity during 2023, followed by a gradual reduction during 2024. A pronounced decline occurred around early 2025, after which discharge volumes remained substantially lower for the remainder of the reporting period. A 7-observation rolling average confirmed that the reduction represented a persistent change in discharge activity rather than a single isolated observation. Potential sudden drops were also identified by comparing discharge volumes with preceding observations. These findings indicate considerable variability in placement activity and a notable reduction in discharge performance during 2025. The observed changes should be evaluated alongside HHS care levels, transfer volumes, and inflow–exit patterns to determine their operational significance.



- **Coefficient of Variation**

$$CV = \frac{Standard\ Deviation}{Mean}$$

A higher CV indicates greater relative variability.

- **Rolling Performance**

A seven-observation rolling average is used to identify changes in recent discharge performance.

Potential sudden drops are flagged when discharge volume falls substantially below the recent rolling baseline.

- **Bottleneck Detection**

Bottlenecks are identified by comparing:

Transfers

vs

Discharges

When transfers consistently exceed discharges, the system experiences **accumulation pressure**.

The dashboard therefore provides threshold-based visual alerts to highlight reporting observations where the selected backlog-pressure threshold is exceeded.

The purpose is to help analysts identify periods requiring closer examination rather than automatically labeling every imbalance as a confirmed operational bottleneck.

9. Streamlit Dashboard

An interactive Streamlit application was developed to translate the analytical results into a monitoring interface.

The dashboard contains four core modules specified in the project requirements.

Module 1 — Care Pipeline Flow Visualization

Displays:

CBP Custody

↓

HHS Care

↓

Sponsor Placement

The dashboard includes an aggregate pipeline-flow visualization and time-series views of CBP and HHS care levels.

Module 2 — Transfer & Discharge Efficiency

Provides:

- Transfer Efficiency Ratio
- Discharge Effectiveness
- Pipeline Throughput
- Time-series efficiency charts
- User-selected ratio toggles

Module 3 — Bottleneck Detection

Provides:

- Net-flow analysis
- Backlog-pressure thresholds
- High accumulation-pressure observations
- Visual alerts

Module 4 — Outcome Trend Analysis

Provides:

- Monthly placement trends
- Month-over-month changes
- Discharge variability
- Outcome Stability Score
- Potential sudden-drop detection

User Capabilities

The dashboard provides three major interactive capabilities.

Date Range Selection

Users can select a specific reporting period for analysis.

Ratio-Based Metric Toggle

Users can switch between:

Transfer Efficiency

Discharge Effectiveness

Both

Threshold-Based Alerts

Users can adjust thresholds for:

- Transfer efficiency
- Discharge effectiveness
- Backlog pressure

The dashboard visually flags observations that cross the selected thresholds.

KPI Framework

The project uses the following KPI framework:

KPI	**Formula**	**Purpose**
Transfer Efficiency Ratio	Transfers ÷ CBP Custody	Evaluate CBP → HHS transfer activity
Discharge Effectiveness	Discharges ÷ HHS Care	Evaluate HHS discharge activity
Pipeline Throughput	Total Exits ÷ Total Entries	Evaluate overall observed pipeline throughput
Backlog Accumulation Rate	Inflow − Successful Exits	Identify accumulation pressure
Outcome Stability Score	Derived consistency indicator	Assess discharge stability

The KPI framework follows the analytical methodology described in the project guide.

Aggregate Reporting Data

The dataset provides aggregate operational counts rather than individual child-level records.

Therefore, the analysis cannot directly calculate:

- individual length of stay,
- individual transition time,
- individual case-processing duration,
- individual reunification time.

Stock vs Flow Measures

CBP custody and HHS care represent populations at a reporting point, whereas transfers and discharges represent movements.

Consequently, the efficiency ratios should be interpreted as **operational indicators** rather than individual transition probabilities.

Reporting-Date Gaps

The observations are not a perfectly continuous daily time series. Therefore, analyses involving consecutive observations should distinguish between:

consecutive reporting observations

and

consecutive calendar days.

No Explicit Reunification-Success Variable

The dataset does not provide a separate variable explicitly labeled reunification_success. Discharges from HHS care are therefore used as the available placement-outcome indicator.

❖ Recommendations

Based on the analytical framework, the project can support several process-oriented recommendations.

1 Monitor Inflow-Exit Imbalance

Regularly compare HHS transfers with discharges to identify periods where inflows consistently exceed successful exits.

2 Use Threshold-Based Early Warning Indicators

Operational teams can establish appropriate thresholds for:

- transfer efficiency,
- discharge effectiveness,
- net-flow accumulation.

3 Monitor Monthly Placement Trends

Month-over-month discharge trends can help identify periods requiring additional investigation.

4 Investigate Sustained Low-Discharge Periods

Periods of prolonged low discharge activity should be reviewed to determine whether they correspond to:

- changes in operational volume,
- reporting changes,
- case-management constraints,
- capacity constraints,
- other process factors.

The dashboard is intended to help identify these periods; the available aggregate dataset alone cannot establish their underlying causes.

5 Use Outcome Stability Monitoring

Rolling performance measures can provide an early indication of unusually large changes in placement activity.

Conclusion

The **Care Transition Efficiency & Placement Outcome Analytics** project provided a structured, data-driven assessment of the UAC care pipeline across three key stages: **CBP Custody → HHS Care → Sponsor Placement**. Using **720 reporting observations covering approximately January 2023 to December 2025**, the analysis examined pipeline volumes, transition efficiency, flow balance, temporal patterns, stagnation periods, and discharge outcome stability.

The analysis demonstrated that **HHS care represents the largest active stage of the observed pipeline**, with an average HHS care level of approximately **6,061.3 children**, compared with an average CBP custody level of approximately **171.5 children**. The aggregate **Transfer Efficiency Ratio was 75.03%**, while **Discharge Effectiveness was 2.86%**, providing operational indicators of movement through the CBP-to-HHS and HHS-to-placement stages.

The backlog analysis showed that recorded successful exits exceeded incoming transfers across the sample observations examined. The cumulative flow-balance measure declined to approximately **−34,000** before stabilizing at lower negative levels. This indicates that the observed transfer-versus-discharge balance did not demonstrate a sustained inflow-driven accumulation pattern. However, because the dataset consists of aggregate reporting observations with gaps between dates, the cumulative measure should be interpreted as a **flow-balance indicator rather than a literal count of unresolved cases**.

Temporal analysis showed relatively similar transfer-efficiency ratios between weekdays and weekends, at approximately **68.8% and 70.0%**, respectively. Discharge effectiveness was approximately **2.27% for weekdays and 2.84% for weekends**. These differences indicate broadly similar operational patterns across the two day types within the available reporting observations.

One of the most significant findings was the substantial change in **discharge and placement activity during 2025**. Across all 720 observations, discharge volume had a mean of **173.41**, a standard deviation of **125.70**, a median of **181**, and a range from **0 to 505** children. Discharge activity was relatively high and highly variable during 2023, gradually declined during 2024, and then experienced a pronounced and persistent reduction around early 2025.

The stagnation analysis identified **six prolonged low-discharge periods during 2025**. The longest extended from **August 24 to December 21, 2025**, covering **82 reporting observations** with an average of **10.35 discharges per observation**. The lowest average discharge level was observed between **March 12 and June 4, 2025**, at approximately **9.22 discharges per observation**. These periods represent important low-activity signals that warrant further examination alongside HHS care levels and inflow volumes.

Overall, the project demonstrates the value of combining **stock measures, flow measures, efficiency ratios, cumulative flow balance, temporal analysis, and outcome-stability indicators** rather than relying on a single KPI. The analysis provides a practical framework for identifying changes in pipeline activity, monitoring potential bottlenecks, and highlighting periods requiring deeper operational investigation.